Design of a Normal Stress Electromagnetic Fast Linear Actuator

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This paper describes the design of a normal stress electromagnetic linear actuator for fast tool servos during nonrotationally symmetric diamond turning. By using the permanent magnet as the biasing flux generator and the total armature pole surface for force generation, the actuator is designed to achieve both linear operation characteristics and high acceleration. A design methodology is presented, which is based on analytical and finite element method magnetic circuit analysis. For design optimization, a new criterion, high actuating force density, is introduced. Based on the optimized structural parameters and the strategy of design for manufacturing, a novel axisymmetric fast linear actuator is developed that has a stroke of 100 μm and 500 G acceleration. The linearity of the actuating force versus both the excitation current and the armature displacement is experimentally demonstrated. It is shown that the experimental and calculated results agree well with each other.

Index Terms—**Fast tool servo, linear actuator, nonrotationally symmetric turning, normal stress.**

I. Introduction

NONROTATIONALLY symmetric turning is a single-point diamond cutting process that generates a workpiece with complicated free-form surfaces by controlling the cutting tool motion in the direction normal to the surface of the workpiece. Such surfaces are used in a wide range of products, including micro-optics, fusion experiment targets, and metal molds for light-enhanced films. Fast tool servo (FTS) plays an important role in this machining process, which activates the diamond tool to complete the reciprocating motion to synchronize with the spindle rotation. Nonrotationally symmetric turning is one of the leading methodologies to fabricate high-quality complicated free-form surfaces [1].

With the increasing complexity of the surfaces requiring more components in shorter spatial wavelength, the FTS must be simultaneously designed with bandwidth up to thousands of Hertz, acceleration greater than hundreds of G, and accuracy around tens of nanometers [2]. This leads to complexity and difficulty in designing and implementing the FTS. As a result, a significant amount of effort has been devoted to the FTS.

To design and implement the FTS, the actuating principle and structure of the linear actuator, as well as the tracking control algorithm for generating accuracy cutting motion are essential and crucial to success. In order to design an FTS with high bandwidth, conventional solutions to the linear actuator are based on the piezoelectric principle, which is the potential to achieve quick response, high acceleration and high stiffness. Dow developed an FTS using a hollow piezoelectric actuator. This FTS has a maximum stroke of 5 μm while operating at 1 kHz, but it could not work continuously due to unacceptably high internal heat [3]. Rasmussen designed a piezoelectric tool servo system for variable depths of cut with a 50 μm stroke and 200 Hz bandwidth [4]. Woronko presented a piezoelectric-based FTS for precision turning of local areas on conventional machine tools. This FTS could provide a 38 μm stroke, 370 N/μm stiffness, 200 Hz bandwidth, and 20 nm tool positioning resolution [5]. Kim described an FTS driven by a piezoelectric actuator for diamond turning of large off-axis aspheric mirrors; the FTS has a 7.5 μm stroke and 100 Hz bandwidth [6]. Gao designed and constructed a FTS with a bandwidth of 2.5 kHz, which employs a piezoelectric tube actuator that actuates the diamond tool to machine a sinusoidal grid surface with spatial wavelengths of 100 μm and amplitudes of 100 nm [7]. In spite of these achievements, there are some inherent drawbacks with piezoelectric actuators, such as nonlinearity, hysteresis, creep, thermal variations, and extension under load. These problems not only increase the difficulty of servo control, but also make the piezoelectric actuator generate large amounts of heat during high frequency operation, which limits the FTS operation at 1 kHz with a few microns stroke [8], [9].

It is known that electromagnetic linear actuators have significantly higher force density in the air gap normal direction than in the shear direction. Such a normal stress electromagnetic actuator has low heat dissipation, and the actuating force is proportional to the current squared and inversely proportional to the air gap squared. As a result, the normal stress actuator potentially gains greater acceleration than shear stress (Lorentz) actuators, such as voice coil motors, and operates over a wider frequency band than the piezoelectric actuators. It is considered to be a promising alternative solution to fast actuators. Although an actuator working on normal stress has a relatively short motion range, such a short stroke is generally acceptable for some FTS applications, such as nonrotationally symmetric diamond turning of micro-optic surfaces. Gutierrez and Ro described a normal stress actuated FTS with a 600 μm stroke and 100 Hz bandwidth [10], [11]. Lu and Trumper presented a normal stress actuated FTS for diamond turning of sinusoidal surfaces. The FTS achieves a 23 kHz closed-loop bandwidth, 1.7 nm RMS positioning error, and 500 G acceleration at 10 kHz [2]. Due to inherent contradictions with respect to stroke, actuating force, acceleration, and bandwidth, a systematic design and optimization methodology for normal stress actuators is required, which has not been addressed in previous investigations.

This paper is motivated by the optimal design problem of normal stress electromagnetic actuators and attempts to present

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a design and optimization method through both analytical derivation and the finite-element method (FEM) in order to fully realize the potential of a normal stress actuator. Moreover, the concept of design for manufacturing is employed to design the mechanical structure of the actuator for convenient machining and assembly.

The rest of this paper is organized as follows. Section II analyzes the normal stress electromagnetic field and actuating force of a basic magnetic circuit through analytical derivation and the FEM. Section III presents the design objectives and the actuator parameter optimization method. Section IV describes the design of the actuator mechanical structure to facilitate easy manufacturing. The experimental results of the actuating force performance are also given in this section. Finally, concluding remarks are given in Section V.

II. Analysis of Normal Stress Electromagnetic Actuating Force

A. Basic Magnetic Circuit of a Normal Stress Actuator

Although normal stress actuation has a high force density, the strong nonlinear behavior of the actuating force will increase the plant complexity and control difficulty significantly; thus, the tracking performance in high frequency operation suffers. Biasing flux can be used to eliminate the nonlinearity by involving permanent magnets in the actuating structure. Fig. 1 shows the actuating principle and configuration of a normal stress electromagnetic actuator using biasing flux [2]. In contrast to typical normal stress actuators [12], the total armature area contributes to force generation, which reduces the inertial load and heat dissipation that result from eddy currents and hysteresis.

As shown in Fig. 1, the actuator is composed of a moving armature, a stator core, a permanent magnet, and excitation coil windings. Both the armature and the stator core are made of soft magnetic materials. Between the permanent magnet and the armature, there is a rubber pad to allow shear motion of the armature in the X-direction relative to the stationary magnet. The air gaps on the left and right side of the armature are denoted as X_0+X and X_0-X, respectively, where X_0 is the air gap when the armature is centered and X is the armature displacement taken as positive for movement to the right. The total excitation coil windings NI are split into two halves on the left and right sides. The permanent magnet and the excitation coil windings generate the total fluxes on the left and right sides of the armature, respectively. The reference direction of both total fluxes is the same as that of X. When the current direction is flowing out of the paper, as shown in the figure, the flux density is enhanced in the right air gap and reduced in the left air gap. The net actuating force is the difference between the electromagnetic forces on the left and right surfaces of the armature. When the direction of the currents that enter the excitation coil windings changes, the direction of the net actuating force also changes, which results in the back and forth motion of the armature.

B. Analytical Approach

1) Magnetic Field Analysis: Lu [2] employed the commonly used magnetic loop analytical approach, assuming that the magnet permeability of the soft magnetic materials is infinite; thus, the reluctance is zero inside both the armature and the stator. However, the working stroke of the actuator is only hundreds of microns. Such a short stroke corresponds to a small

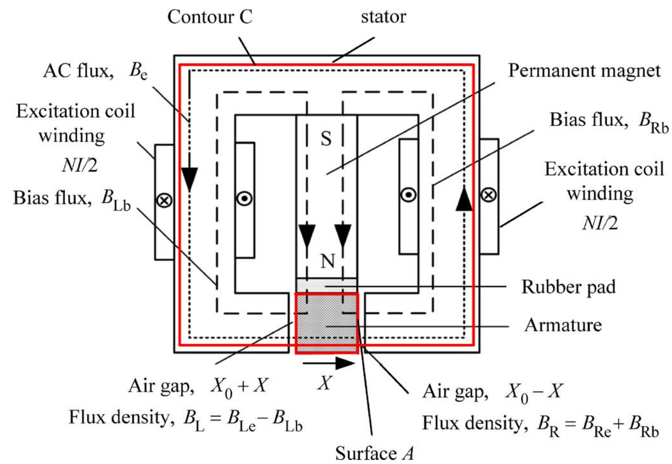

Fig. 1. Actuating principle and magnetic circuit of the normal stress electromagnetic actuator.

air gap, which results in small air gap reluctance. Therefore, the relatively small reluctance inside the armature and the stator core must be involved in the analysis to improve computation accuracy.

As shown in Fig. 1, the total fluxes, B_{L} and B_{R}, in the air gap contain the biasing components, B_{Lb} and B_{Rb} generated by the permanent magnet and the AC components, B_{Le} and B_{Re} generated by the excitation coil windings. In this section, two flux components will be analyzed in the biasing magnetic circuit and the excitation magnetic circuit separately.

The biasing magnetic circuit is analyzed first, assuming that the excitation coil current is zero. As illustrated in Fig. 1, the biasing flux enters the armature from the top surface and splits inside the armature; then, it exits the armature into the left and right air gap, respectively. The total flux flowing from the permanent magnet into the armature is denoted as Φ_{arm}. Applying the Gauss Law on the surface A in Fig. 1, the following equation is derived:

$$(B_{\mathrm{Lb}}+B_{\mathrm{Rb}})S_{\mathrm{arm}}-\Phi_{\mathrm{arm}}=0 \tag{1}$$

where $S_{\mathrm{arm}}$is the armature pole face area at each working air gap. Next, the Ampere loop Law is applied to the contour C in Fig. 1 to obtain

$$\oint_C \vec{H}\cdot \mathrm{d}\vec{l}=0$$

$$B_{\mathrm{Rb}}\left[(X_0-X)+\frac{L_{\mathrm{sta}}}{2\mu_{\mathrm{r}}}\right]-B_{\mathrm{Lb}}\left[(X_0+X)+\frac{L_{\mathrm{sta}}}{2\mu_{\mathrm{r}}}\right]=0 \tag{2}$$

where L_{sta} is the stator core length along the magnetic circuit direction, and μ_r is the relative magnetic permeability inside the stator core of soft magnetic material.

Combining (1) and (2), the following is derived:

$$\begin{cases} B_{\mathrm{Rb}}=B_{\mathrm{arm}}+\dfrac{\mu_{\mathrm{r}}B_{\mathrm{arm}}}{\mu_{\mathrm{r}}X_0+L_{\mathrm{sta}}/2}X \\ B_{\mathrm{Lb}}=-\left(B_{\mathrm{arm}}-\dfrac{\mu_{\mathrm{r}}B_{\mathrm{arm}}}{\mu_{\mathrm{r}}X_0+L_{\mathrm{sta}}/2}X\right) \\ B_{\mathrm{arm}}=\dfrac{\Phi_{\mathrm{arm}}}{S_{\mathrm{arm}}}=\alpha_{\mathrm{lea}}B_r \end{cases} \tag{3}$$

where B_{arm} is denoted as the flux density flowing from the permanent magnet into the armature, B_r is the remanence of the permanent magnet, and α_{lea} is the leakage coefficient that illustrates the leakage of the magnet circuit quantitatively. Once the magnet circuit is designed, α_{lea} is usually constant. Therefore, assuming that B_{arm} is constant, the biasing flux densities B_{Lb} and B_{Rb} are linear functions of the armature displacement X according to (3).

While the AC flux generation is investigated, it is assumed that the permanent magnet has been removed. As shown in Fig. 1, there is only one closed circuit relating to the AC flux. The left and right coil windings generate flux densities B_{Le} and B_{Re}, respectively. Applying Gauss's Law to surface A in Fig. 1, the following is derived:

$$B_{\mathrm{Le}}S_{\mathrm{arm}} - B_{\mathrm{Re}}S_{\mathrm{arm}} = 0 \tag{4}$$

which illustrates that the AC flux densities of B_{Le} and B_{Re} are equal. Next, applying Ampere's Loop Law to the contour C in Fig. 1 the following is obtained:

$$\frac{B_{\mathrm{Le}}}{\mu_0}\left(X_0 + X + \frac{L_{\mathrm{arm}}}{2\mu_{\mathrm{r}}} + \frac{L_{\mathrm{sta}}}{2\mu_{\mathrm{r}}}\right) + \frac{B_{\mathrm{Re}}}{\mu_0}\left(X_0 - X + \frac{L_{\mathrm{arm}}}{2\mu_{\mathrm{r}}} + \frac{L_{\mathrm{sta}}}{2\mu_{\mathrm{r}}}\right) = NI \tag{5}$$

where L_{arm} represents the excitation magnetic circuit length in the armature range, NI is the number of Ampere-turns of the total coil windings, and μ_0 is the vacuum permeability.

Setting $L_{\mathrm{fer}} = L_{\mathrm{arm}} + L_{\mathrm{sta}}$ and simplifying (5), the AC fluxes in both air gaps can be expressed as

$$B_{\mathrm{Le}} = B_{\mathrm{Re}} = \frac{\mu_0\mu_{\mathrm{r}}}{2\mu_{\mathrm{r}}X_0 + L_{\mathrm{fer}}}NI. \tag{6}$$

This indicates that the AC flux densities are a linear functions of the excitation current I and independent of the armature displacement X. After superposition of the biasing flux in (3) and the AC flux in (6), the total fluxes in the left and right air gaps can be described as

$$\begin{cases} B_{\mathrm{L}} = \dfrac{\mu_0\mu_{\mathrm{r}}}{(2\mu_{\mathrm{r}}X_0 + L_{\mathrm{fer}})}NI - \left(B_{\mathrm{arm}} - \dfrac{\mu_{\mathrm{r}}B_{\mathrm{arm}}}{\mu_{\mathrm{r}}X_0 + L_{\mathrm{sta}}/2}X\right) \\ B_{\mathrm{R}} = \dfrac{\mu_0\mu_{\mathrm{r}}}{(2\mu_{\mathrm{r}}X_0 + L_{\mathrm{fer}})}NI + \left(B_{\mathrm{arm}} + \dfrac{\mu_{\mathrm{r}}B_{\mathrm{arm}}}{\mu_{\mathrm{r}}X_0 + L_{\mathrm{sta}}/2}X\right) \\ B_{\mathrm{arm}} = \alpha_{\mathrm{lea}}B_{\mathrm{r}} \end{cases}. \tag{7}$$

The reference direction for B_{L} and B_{R} is taken as positive in the X-direction.

2) Actuating Force Analysis: In this section, Maxwell stress tensor theory is used to calculate the net actuating force on the armature. As shown in Fig. 1, the actuating force is the vector sum of the normal Maxwell force actuating on the left and right pole areas of the armature. Further, assuming the air gap flux densities B_{L} and B_{R} are normal to the left and right pole areas of the armature and evenly distributed, the electromagnetic forces acting on the left and right pole areas of the armature are calculated by applying the Maxwell stress tensor theory

$$F_{\mathrm{L}} = \frac{B_{\mathrm{L}}^2 S_{\mathrm{arm}}}{2\mu_0}, \quad F_{\mathrm{R}} = \frac{B_{\mathrm{R}}^2 S_{\mathrm{arm}}}{2\mu_0}. \tag{8}$$

Substituting (7) into (8) leads to

$$\begin{cases} F_{\mathrm{L}} = \dfrac{S_{\mathrm{arm}}}{2\mu_0}\left[\dfrac{\mu_0\mu_{\mathrm{r}}}{(2\mu_{\mathrm{r}}X_0 + L_{\mathrm{fer}})}NI - \left(B_{\mathrm{arm}} - \dfrac{\mu_{\mathrm{r}}B_{\mathrm{arm}}}{\mu_{\mathrm{r}}X_0 + L_{\mathrm{sta}}/2}X\right)\right]^2 \\ F_{\mathrm{R}} = \dfrac{S_{\mathrm{arm}}}{2\mu_0}\left[\dfrac{\mu_0\mu_{\mathrm{r}}}{(2\mu_{\mathrm{r}}X_0 + L_{\mathrm{fer}})}NI + \left(B_{\mathrm{arm}} + \dfrac{\mu_{\mathrm{r}}B_{\mathrm{arm}}}{\mu_{\mathrm{r}}X_0 + L_{\mathrm{sta}}/2}X\right)\right]^2 \end{cases}. \tag{9}$$

The net actuating force on the armature is thus calculated and given by

$$F = F_R - F_L = \frac{2\mu_{\mathrm{r}}S_{\mathrm{arm}}B_{\mathrm{arm}}}{2\mu_{\mathrm{r}}X_0 + L_{\mathrm{fer}}}NI + \frac{4\mu_{\mathrm{r}}S_{\mathrm{arm}}B_{\mathrm{arm}}^2}{2\mu_0\mu_{\mathrm{r}}X_0 + \mu_0 L_{\mathrm{sta}}}X. \tag{10}$$

Note that the coefficients before NI and X are constants relating to the actuator geometry and material. Equation (10) can be further expressed as

$$F = k_i I + k_x X \tag{11}$$

where k_i and k_x are constant coefficients. From (11), it is observed that the actuator is linear in principle because the net actuating force is linear functions of both the excitation current I and the armature displacement of X. The component $k_i I$ describes the desired actuating force generated by the excitation coil windings. The component $k_x X$ is an accessional flux force when the armature departures from the centered position, which should be rejected in future servo control of the actuator.

C. Numerical Calculation Using FEM

The aforementioned analytical approach provides a conduit to invaluable insights on how and why the normal stress electromagnetic actuator works. However, if an accurate computation is required, there are some limitations and drawbacks with the analytical approach. First, the air gap flux densities B_{L} and B_{R} are assumed to be constant and normal to the armature pole area. Then, for simplification, the fixed relative magnet permeability is used to describe the soft magnetic material characteristic. In fact, the B-H characteristic of the soft magnetic material is nonlinear, which will introduce errors in strong or high frequency magnetic fields. Finally, it is difficult to obtain an accurate leakage coefficient of α_{lea} in the analytical calculation. Usually, the difficulty is solved with the aid of design experience and handbooks. In this section, an accurate modeling and analysis of the electromagnetic field of the actuator is achieved by employing a FEM package in the Cartesian coordinates. The FEM is a numerical method, which solves the Maxwell equations numerically.

1) Simulation of the Flux Density Distribution: In this study, we developed an electromagnetic field analysis program that takes into account the combined effect of the permanent magnet and the excitation coil windings using the ANSYS 10.0 software package. For the field analysis, initially, the B-H curve of the soft magnetic material is stored in the FEM program. Then, an automatic meshing generation procedure is used, and the three-dimensional vector magnetic flux densities in various parts of the magnetic circuit are computed. Because the flux density in the X-direction contributes to the actuating force generation, and it is far larger than those of the Y- and Z-directions according to theoretical analysis [13], only the air gap flux distribution along the X-direction is investigated.

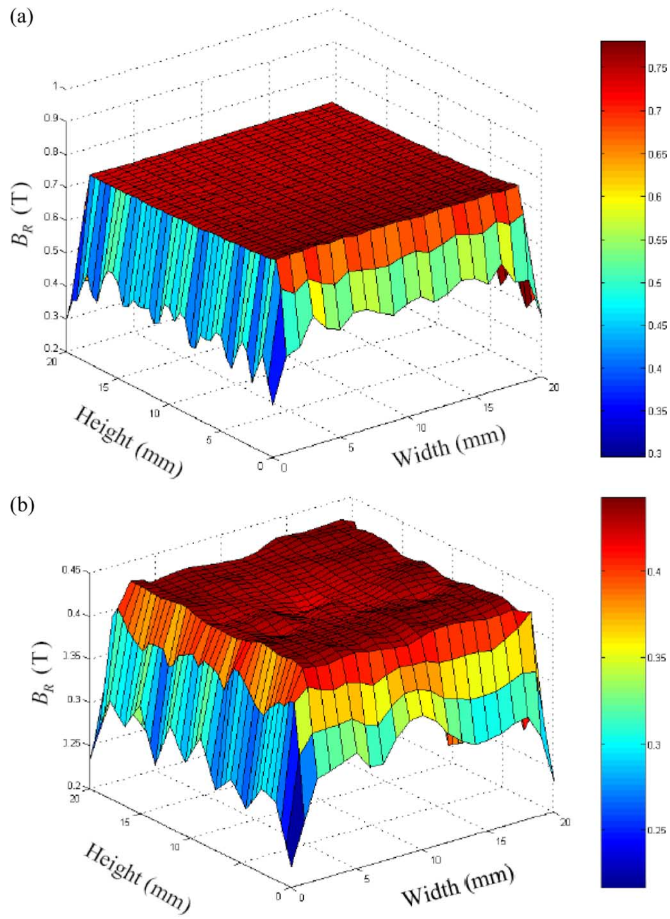

Fig. 2. Distribution of the flux density $B_{\rm R}$ (a) $X_0 = 0.1$ mm, (b) $X_0 = 1$ mm.

A typical armature that is a cube with lengths of 20 mm is defined. This implies that the pole area of both the armature and the stator core are squares with lengths of 20 mm. The permanent magnet material is Nd-Fe-B, and the soft magnetic material for the armature and the stator core is 1J50. The Ampere turns, NI, is set to 72 At. For illustration, one cross section in the right air gap is selected for calculation. The cross section is 5 μm away from the right pole area of the armature. Fig. 2 shows the flux density distribution of the selected cross section, in which two cases, $X_0 = 0.1$ mm and $X_0 = 1$ mm, are analyzed.

As illustrated in Fig. 2(a), when the centered air gap X_0 is 0.1 mm, the flux density $B_{\rm R}$ distribution is highly uniform, ranging from 0.72 T to 0.76 T, except the air gap edge. If the edge flux density is considered carefully, a significant decrease is observed. However, the area with decreased flux density is only 10 percent of the total air gap area, which will not significantly reduce the actuating force. Moreover, such a variant flux density distribution is symmetrical to the armature center. This will not generate a rotational moment.

When X_0 increases to 1.0 mm, an uneven flux density distribution can be observed in Fig. 2(b) after comparison with Fig. 2(a). Furthermore, the flux density in the working area is reduced to 0.40 T to 0.43 T. This indicates that a smaller value of X_0 is desired to obtain high flux density and even distribution. But, it will make it difficult to manufacture the actuator. There is a tradeoff between the electromagnetic performance and the manufacturing cost of the actuator.

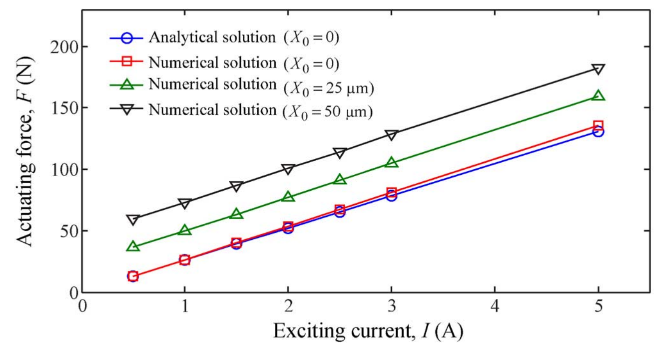

Fig. 3. Analytical and numerical solutions of the actuating force.

2) Numerical Calculation of Actuating Force: A professional electromagnetic FEM software package, Maxwell 2-D/3-D, is used to calculate the actuating force. To improve computation accuracy, some modeling experiences are proved to be valid. First, the flux sources of the excitation coil current and the permanent magnet should be set separately. Secondly, for the RZ rotationally symmetric model, both $\rm R_\infty$ and $\rm Z_\infty$ should be taken as balloon boundaries to reduce the calculation time. Then, due to the small air gap in the X direction, the mesh should be completed manually to improve computation accuracy in this space. Finally, an adaptive iteration method is employed.

Fig. 3 compares the analytical and numerical results, in which the parameters of the actuator are the same as those in Fig. 2. Notice that the actuating force is linear functions of both the excitation current and the armature displacement. There is little difference between the analytical and numerical results under the same conditions. This is mainly due to the fact that the flux density in the air gap is assumed to be identical in the analytical approach. Therefore, the analytical expressions for the flux density and actuating force, as described in (7) and (10), can be used for preliminary actuator design. On the other hand, such little difference demonstrates that the effects of the magnetic permeability of the soft magnetic materials on the magnetic fields should be taken into account while analyzing the electromagnetic field.

III. Optimum Design Study

From (10), it can be noted that the actuating force is influenced by the magnet circuit geometry and the materials performances. The centered air gap X_0 and the armature dimensions are two important design parameters. With a smaller X_0, the actuating force will increase. However, a very small air gap results in a significant manufacturing difficulty. On the other hand, larger armature pole area is required to obtain a greater actuating force. However, an increased pole area causes a remarkable increase in the mass of the moving armature, and thus reduces the acceleration capacity of the actuator. Therefore, an optimization study is needed for the air gap and the armature dimensions by using FEM-based numerical calculation [14].

A. Axisymmetric Magnetic Circuit Configuration

Since the designed normal stress actuator does not require the middle part of the armature in the flux path, the actuator can easily be paralleled with multiple armatures. One unique way to achieve this is to design the armature to be a rotational shape.

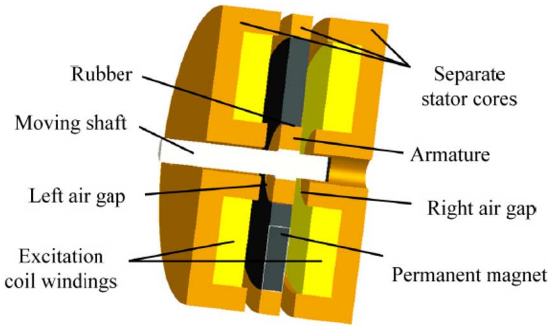

Fig. 4. Cross section of the axisymmetric normal stress electromagnetic actuator.

Based on the design concept, this paper proposes a novel axisymmetric magnetic circuit configuration, as shown in Fig. 4. It can easily be seen that all components are designed to be rotational and the pole area in the air gap becomes annular instead of a square. Since the total annular area of the armature contributes to force generation, a higher actuating force can be obtained. Moreover, such an axisymmetric configuration has a low actuator manufacturing cost.

B. *Optimum Design Objectives*

The goal of the design is to demonstrate the feasibility of a fast tool servo design with high acceleration and realize the potential of normal stress actuators. Therefore, two optimum objectives are designed. On the one hand, a maximum actuating force F in the X direction is required for high dynamic stiffness of the actuator, while maintaining the same actuator geometries and actuating force linearity. On the other hand, a maximum actuating force density is desired. The actuating force density is defined as

$$P_f = \frac{F}{V_{\mathrm{arm}}} \tag{12}$$

where V_{arm} is the volume of the armature. Obviously, P_f reflects the specific actuating force for a unit volume of the moving armature, which leads to smaller actuator dimensions and greater actuator acceleration.

C. *Optimum Design of Parameters*

The analytical expression of the actuating force in (10) illustrates that the geometries of the armature and the air gap significantly affect the actuator performance. Therefore, the main design parameters include the centered air gap, X_0, the external and internal radii of the armature, R_{arm} and r_{arm}, respectively, and the axial thickness of the armature, T_{arm}. According to engineering experience [15], the initial values are set as follows: $X_0 = 0.2$ mm, $R_{\mathrm{arm}} = 15$ mm, $r_{\mathrm{arm}} = 5$ mm, and $T_{\mathrm{arm}} = 10$ mm.

Further, the permanent magnet is made of high energy Nd-Fe-B magnetic material whose type is N38H with 1.22–1.26 T remanence. According to engineering experience [15], the difference between the external and internal radii of the permanent magnet is 29.5 mm. Both the armature and the stator core are made of 1J50, which is a type of widely-used un-oriented soft magnetic material. On the other hand, the excitation coil turns is changed from previous 24 to current 100, while maintaining the previous total number ampere turns. This helps to make full use of the excitation coil winding space and reduce the excitation current and, thus, heat dissipation. To avoid saturating the soft magnetic material and creating too large of an excitation current, the current is set to ± 1 A.

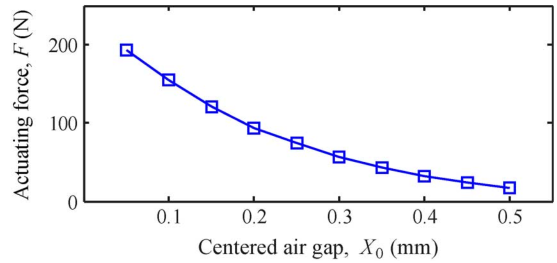

Fig. 5. Effect of the centered air gap, X_0 on the actuating force.

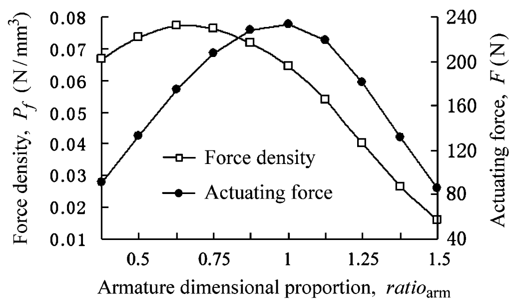

Fig. 6. Effect of the axial thickness of the armature on both the actuating force and the force density.

1) Centered Working Air Gap X_0: The design of X_0 should make the actuating force as high as possible to generate high and even air gap flux density distributions while avoiding flux saturation. Fig. 5 shows the numerical relationship between the actuating force and X_0. Note that a smaller X_0 results in a higher actuating force. In particular, when X_0 is reduced from 0.2 mm to 0.1 mm, an increase of 38.5 percent of the actuating force is obtained. However, a further reduction of X_0 will increase manufacturing difficulty. Therefore, X_0 is designed to be 0.1 mm to easily ensure an actuator with a 50 μm stroke.

2) Optimum Design of the Axial Thickness of the Armature: In this section, a new parameter is first defined as

$$ratio_{\mathrm{arm}} = \frac{T_{\mathrm{arm}}}{R_{\mathrm{arm}} - r_{\mathrm{arm}}}, \tag{13}$$

which describes the dimension proportion between the axial and radial structure of the armature. In a typical normal stress actuator [2], $ratio_{\mathrm{arm}}$ is designed to be 1, which means that the axial cross section of the armature is a square. However, whether there is an optimum $ratio_{\mathrm{arm}}$ that leads to higher force density and actuating force should be investigated.

Fig. 6 shows the effects of the armature dimensional proportion on the force density and the actuating force, in which $R_{\mathrm{arm}} - r_{\mathrm{arm}}$ equals 8 mm. As illustrated in Fig. 6, when $ratio_{\mathrm{arm}}$ is larger than 1, both the actuating force and the force density decrease as X_0 increases. In addition, when $ratio_{\mathrm{arm}}$ is 1, a maximum actuating force is obtained, and when $ratio_{\mathrm{arm}}$ is 0.625, a maximum force density is obtained. Therefore, to achieve both a high actuating force and force density, $ratio_{\mathrm{arm}}$ should be around 0.8125, which is the mean of 0.625 and 1.

3) Selection of the Size Type: The pole area size of the armature is a key parameter that significantly influences both the actuating force and the force density. For the developed actuator with a rotational configuration, the pole area of the armature is

TABLE I
COMPARISONS OF THE ACTUATOR GEOMETRY AND SPECIFICATIONS

Symbol	Quantity	Initial design	Optimized result
X_0	Centered air gap	0.2 mm	0.1 mm
R_{arm}	External radius of armature	15 mm	11 mm
r_{arm}	Internal radius of armature	5 mm	5 mm
T_{arm}	Armature thickness	10 mm	5 mm
F	Maximum actuating force	247 N	155 N
m	Armature mass	51.52 g	12.52 g
P_f	Maximum force density	0.03931 N/mm^3	0.10561 N/mm^3

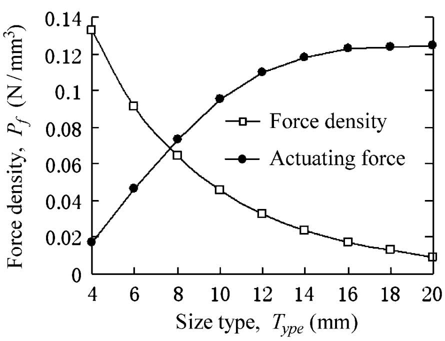

Fig. 7. Effects of the size type on the actuating force and force density.

related to the difference between the maximum and minimum radii, namely $R_{\text{arm}} - r_{\text{arm}}$. Therefore, the actuator can be described by a new characteristic parameter

$$T_{\text{ype}} = R_{\text{arm}} - r_{\text{arm}}, \tag{14}$$

where T_{ype} is the size type of the actuator. Obviously, a larger value of T_{ype} increases the actuating force, but decreases the force density as a result of the moving armature mass. An optimal design of T_{ype} is required. Fig. 7 illustrates the effects of T_{ype} on both the actuating force and the force density. In this figure, X_0 is 0.1 mm, and the armature thickness, T_{arm}, is determined according to the optimized $ratio_{\text{arm}}$. The excitation current varies with the size type to ensure the total flux density in the soft magnetic material approaches saturation. As illustrated in Fig. 7, as T_{ype} increases, the actuating force increases but the force density decreases gradually. Since the developed FTS is designed for diamond turning of non-rotationally symmetric workpieces, greater acceleration is more important than a high actuating force. In diamond turning, the cutting force is small. From this view, it is better to select T_{ype} to be 6 mm, which not only achieves high force density but also does not make the actuating force too small.

4) Optimum Design Results: Table I summarizes comparisons of the actuator geometric dimensions and specifications between the initial design and optimized design.

From Table I, it is noted that the optimized geometric dimensions of the actuator are smaller than those in the initial design. A greater force density is achieved while maintaining the desired maximum actuating force, which improves the actuator acceleration and has the capacity to reject cutting force. The coefficient between the actuating force and the excitation current, $k_{i\_theory} = 155\ \text{N/A}$.

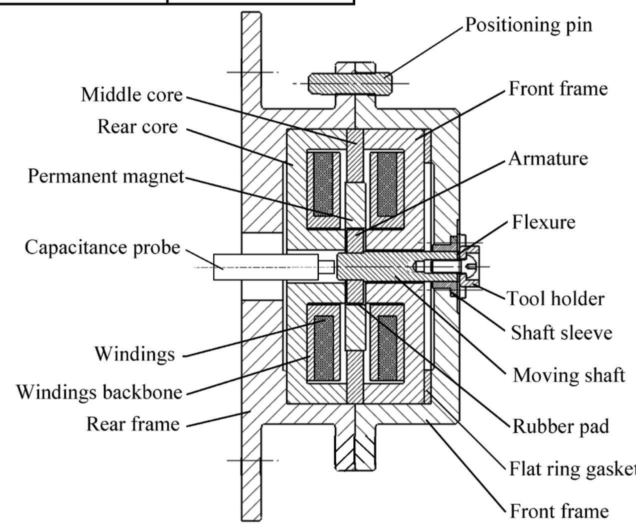

Fig. 8. Schematic assembly configuration of the actuator.

IV. IMPLEMENTATION AND EXPERIMENTS

Based on the optimized design results, a prototype of the axisymmetric normal stress electromagnetic actuator is specially developed. The actuating performances of the actuator are also measured and demonstrated.

A. *Structure Design of the Actuator*

The schematic design of the proposed normal stress actuator is shown in Fig. 8, which is built upon the actuating principle and configuration in Fig. 4 by including support bearings and convenient assembly.

As shown in Fig. 8, the motion of the armature is transferred to the tool holder by a shaft. There is interference between the internal hole of the rotational armature and the shaft to ensure reliable motion transmission. In addition, there is a step surface on the shaft to provide the axial position of the moving assembly. A rotational sleeve is designed to make the shaft center around the armature axis. The total moving assembly is suspended by a spring steel flexural bearing in the front of the moving assembly and a rubber pad bearing in the rear of the moving assembly. Consequently, the moving assembly moves primarily along the axial direction of the armature and the other five degrees of freedom are constrained.

In this design, the stator core is made of 1J50 soft magnetic material. It is split into three parts to simplify manufacturing: the front core, the middle core and the rear core. Also, the surrounding frame is designed to be composed of two parts to make manufacturing convenient: the front frame and the rear frame. They are made of non-magnetic stainless steel and provide a

Fig. 9. Picture of the developed actuator prototype.

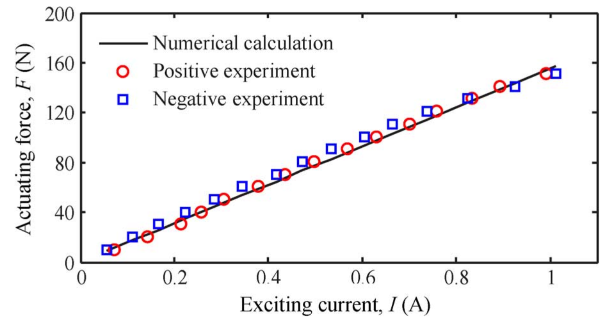

Fig. 10. Numerically calculated and experimentally measured actuating forces versus excitation current.

rigid base for the actuator. The rotational Nd-Fe-B permanent magnet is assembled with eighteen sectors that are magnetized along the radial direction. High strength epoxy polyurethane is used to bond the middle core and the permanent magnet. The excitation coil windings are separated into two parts and wound on two winding backbones.

In the actuator design, the initial air gap is the most important parameter. Generally, to ensure identical initial air gaps on the left and right sides of the armature, the dimensions of the components relating with the air gaps must be designed and manufactured with high accuracy. This will increase manufacturing difficulty. Therefore, a flat ring gasket is specially designed between the front core and the front frame. In this design concept, the axial dimensions of the frames and the stator cores are designed with economic accuracy. The difficulty of maintaining identical initial air gaps on the left and right sides of the armature can be solved by changing the thickness of the flat ring gasket. In addition, a capacitance probe (ADE 5508) with a 100 kHz bandwidth is located at the rear of the moving shaft to measure its motion directly.

Fig. 9 shows a picture of the developed normal stress electromagnetic linear actuator prototype.

B. Experimental Results

The experiments are conducted to demonstrate the potential of the normal stress actuator to realize high acceleration. Moreover, the design concept of linearizing the actuating force by biasing the flux is demonstrated.

The analytical and numerically calculated results shown in Fig. 3 illustrate that the actuating force should be linear functions of both the excitation current and the armature displacement. Therefore, two experiments are designed for the demonstration, respectively.

The first experiment is to measure the functional relationship between the actuating force and the excitation current while maintaining the moving shaft in a fixed and balanced position. A regulated DC power supply is used to provide a variable excitation current, and a force gauge is used to push against the moving shaft. When the direction of the excitation current is changed, the armature moves in the opposite direction. By recording the power supply current and the gauge outputs, the static coefficients between the actuating force and excitation current in two directions can be calculated. In this experiment, to maintain a constant armature position and eliminate the effect of armature displacement on the actuating force, the capacitance probe is used to detect and monitor the armature position in real-time.

Fig. 10 compares the experimentally measured and FEM-based numerically calculated results. It is remarkable that good agreements are observed between the measured and calculated results, as well as between the positive and negative directional results. By fitting the experimental data, we can obtain the proportional coefficients between the actuating force and the excitation current: 153.0 N/A in the positive direction and 150.2 N/A in the negative direction. The resulting force coefficients are very close to the numerical calculation value, $k_{\rm i\_theory} = 155\ \mathrm{N/A}$. This demonstrates that the actuating force of the actuator is a linear function of the excitation current. This is very encouraging because the control of the actuator is very straightforward.

The second experiment is to measure the functional relationship between the actuating force and the moving assembly displacement without the excitation current. In this experiment, the force gauge is used to push against the moving shaft, and the capacitance probe is used to detect the position of the moving shaft when it is stable and motionless. By recoding the gauge outputs and the capacitance probe outputs, the coefficient between the actuating force and the armature displacement can be calculated. Because the moving assembly is supported by flexure, the resulting force coefficients reflect the combined effect of the spring force and actuating force without the excitation current.

Fig. 11 shows the experimentally measured results in the positive and negative directions. It is observed that the actuating forces in the two directions are linear with the armature displacement, and the fitted proportional coefficients are 4.4322 $\mathrm{N}/\mu\mathrm{m}$ for the positive direction and 3.4996 $\mathrm{N}/\mu\mathrm{m}$ for the negative direction. This is because the spring coefficients in the two directions are slightly different as a result of assembly. Although the difference makes the actuator behave differently in the two directions, it will be estimated and compensated for by the servo control algorithm to achieve the desired closed-loop performance.

Finally, the developed normal stress actuator prototype has been tested in an open loop to measure the acceleration capacity. The actuator runs in a sinusoidal trajectory that has an amplitude of 1.5 μm and frequency of 10 kHz. This implies that the actuator achieves at least 500 G acceleration.

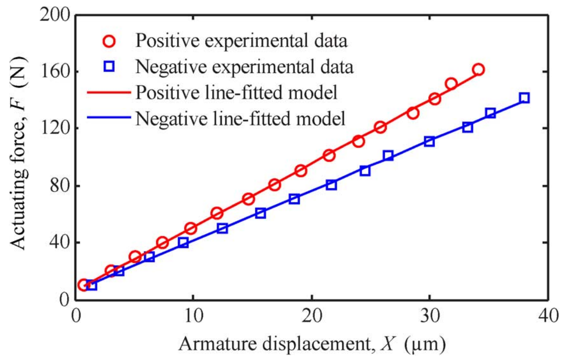

Fig. 11. Experimentally measured actuating force versus the armature displacement.

V. Conclusion

This paper presents the analytical derivation, numerical calculation, design optimization, implementation, and experiments for a novel axisymmetric normal stress electromagnetic fast linear actuator. The results show that the fast actuator is very promising for a precision tool servo system with short stroke and high bandwidth. It effectively linearizes the actuating force by introducing the biasing flux into the magnetic circuit using the permanent magnets. For the analytical analysis of the electromagnetic field, the magnet permeability of the armature and stator core made by the soft magnetic materials must be taken into account to achieve an accurate analytical expression. Moreover, the proposed optimal design method benefits the realization of high acceleration actuator by exploiting the actuator potential. The unique axisymmetric configuration and the centered air gap adjustment approach make the actuator manufacture conveniently. Future efforts will focus on closed-loop control design of the actuator to compensate for its dynamic uncertainty and thus achieve better tracking and more robust performance.

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